UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION 1 (5 marks)

Question 1 is based on the following statement.
Mr. X is known as Father of Computer. He developed Analytical Machine in 1837 which has four main components, namely; memory, the mill, punch card reader and the punch and printed section. Analytical Machine is the successor of Y machine. Y machine is used to compute tables of numbers for naval navigation.

a) Name the inventor of Mr. X.

b) Identify the name of Y machine.

c) What is the function of Analytical Machine?

d) Map the components Analytical Machine to Von Neumann architecture

QUESTION 2 (5 marks)

a) How classification of parallel processor is constructed?

b) Give ONE (1) example for each parallel processor architecture.

QUESTION 3 (5 marks)

The advancement of technology through different computer generations has focused on improving performance, speed, and efficiency. From vacuum tubes in the first generation to modern AI-powered systems, each generation introduced faster processing, smaller components, and greater reliability.

a) Name the inventor of the following devices:

   i. Mechanical Calculating Machine

   ii. Analytical Machine
   iiD
   IAS Machine

b) Why are the machines of later computer generation designed for general purposes instead
of the eadier computer generation that is more specific in function?

QUESTION 4 (5 marks)

a) \Mat is a bit?

b) Given a system of vacuum tubes where each tube represents a single bit with two states: ON=1 and OFF=O. Identify the number of vacuum tubes needed to store the value of gro. Show your workflow to justify your answer.

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i. Binary system.

ii. Octal system.

QUESTION 5 (10 marks)

a) Discuss TWO (2) functions of an input/output module.

b) External input/output devices can be broadly categorized into three categories. List TWO (2) of the categories and provide an appropriate example of devices for each categories.

c) Name ONE (1) input/output technique that requires active CPU intervention and ONE (1) technique that does not require CPU intervention.

QUESTION 6 (10 marks)

a) Identify the correct access method based on the following scenario
Access method
Scenario
Readi
I
files.
c file from a folder.
Accessi
a
Streami
files line
line.
Accessin
AS ecific file block usi
an index in a la
e file.
toas
cific frame or a timestam
in the video.
Jum
tn

b) Rank the following memory types based on the most to the least storage capacity' Main memory, Tape, LI Cache, Hard drive, ReBister, L3 Cache

c) Discuss the term volatile from the perspective of computer memory.

QUESTION 7 (10 marks)

a) Convert -81.21875 to a floating point number in IEEE single precision format in hexadecimal value form. Show your complete workflow.

b) Convert 43A37C00h in IEEE single precision floating point number to decimal value form.

Show your workflow.

QUESTION 8 (10 marks)

Question 7 is based on Figure 1.
Define intenupt.

a) Identify the intemrpt that has the highest priority with relevant justification

b)

c) Identify the interrupt that has the lowest priority with relevant justification

d) Rank the sequence of the intem.rpt process completed ff the computer system applies the
defined priority concept to handle multiple intemrpts.

e) Rank the sequence of the interrupt process completed if the co
mputer system aPPlies the
disable nested interrupt concept to handle multiple interrupts'

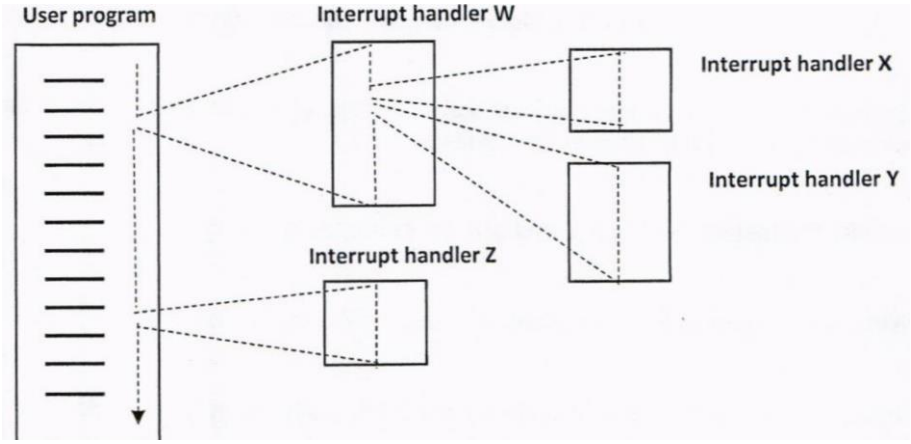


Figure 1: Nested interrupts

QUESTION 9 (10 marks)

a) When is an I/O module installed?

b) What will happen if the I/O module is not installed?

c) Explain ONE (1) category of I/O module

d) What does it mean by 'hot connection'?

e) List THREE (3) common interfaces to external devices.

QUESTION 10 (10 marks)

a) Discuss the main function of cache memory.

b) How can the data stored in the cache and main memory be synchronized.

c) Contrast between SRAM and DRAM based on the following characteristics:
Type of storage
ii,

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Availability of charges leak
price
Example of location of memory usage

PART B (20 MARKS)

Answer the question in this part.

QUESTION 11 (20 marks)

Question 1 is based on Figure 2 and Table 1

Write an appropriate program using conect assembly language notations to compute the equation shown in Figure 2 using instructions for zero-address, one-address, and three-address. The available instructions are provided in Table 1. The registers labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register z.

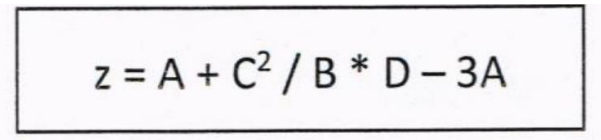


Figure 2

Table 1: It/achine instruction

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	SUB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	

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